



# Interpretable Machine Learning

A Guide for Making Black Box Models Explainable

Author

Christoph Molnar

**PRESENTED BY: MUHAMMAD ATIF SAEED**

# Interpretability

**Interpretability** is the degree to which a human can understand the cause of a decision.

A method is **interpretable** if a user can correctly and efficiently predict the method's results

**Interpretability** is about mapping an abstract concept from the models into an understandable form.

**Explainability** is a stronger term requiring interpretability and additional context.

Interpretability is debugging tool for detecting bias in ML Model.  
Daily lives requires interpretability to increase social acceptance.  
Explanations are used to manage social interactions.

# XAI traits

Fairness: Ensuring that predictions are unbiased and don't implicitly or explicitly discriminate against underrepresented groups.

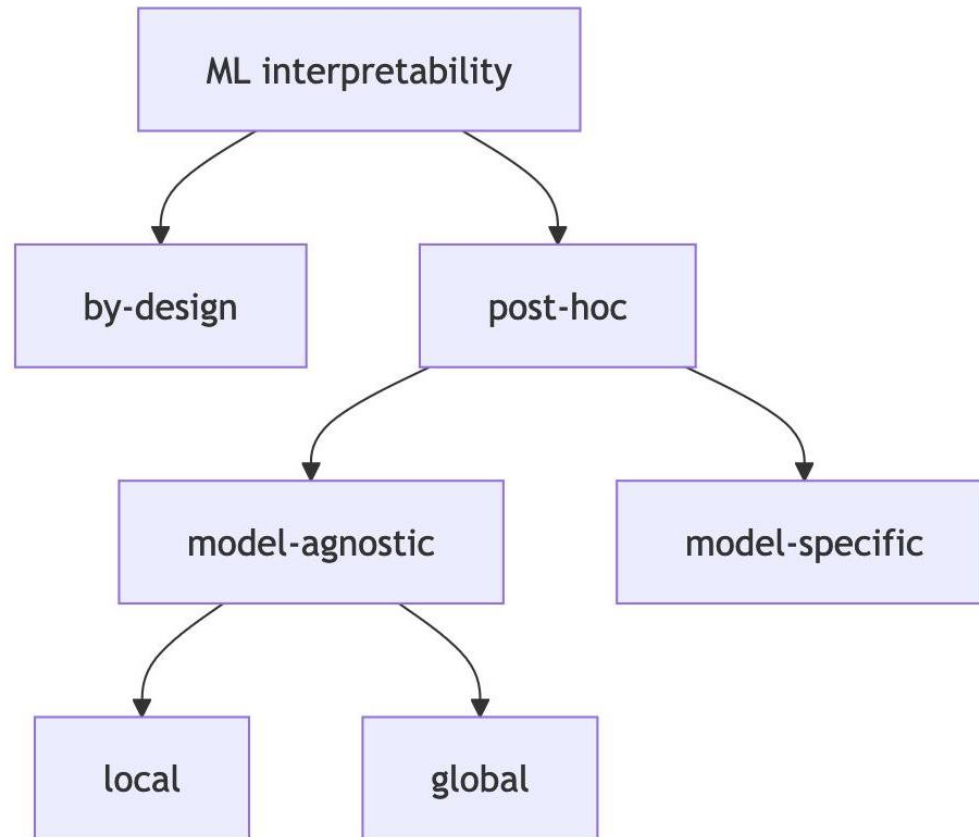
Privacy: Ensuring that sensitive information in the data is protected.

Reliability or Robustness: Ensuring that small changes in the input don't lead to large changes in the prediction.

Causality: Ensure that only causal relationships are picked up.

Trust: It's easier for humans to trust a system that explains its decisions compared to a black box.

# Goals of Interpretability



# Interpretable models by design

Linear regression

Fit a linear model by minimizing the sum of squared errors.

Logistic regression

Extend linear regression for classification using a nonlinear transformation.

Linear model  
extensions

Add penalties, interactions, and nonlinear terms for more flexibility.

Decision trees

Recursively split data to create tree-based models.

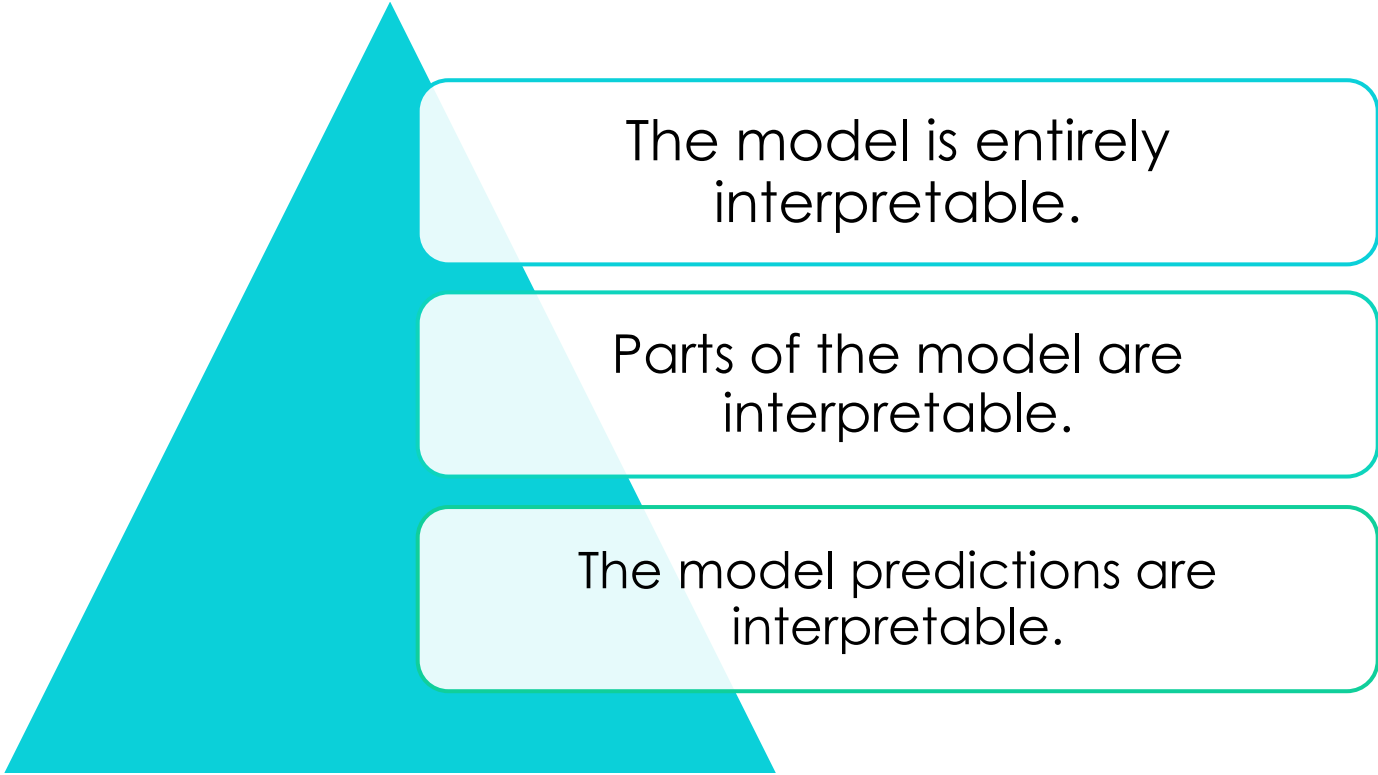
Decision rules

Extract if-then rules from data.

RuleFit

Combine tree-based rules with Lasso regression to learn sparse rule-based models.

# Scope of interpretability



The model is entirely interpretable.

Parts of the model are interpretable.

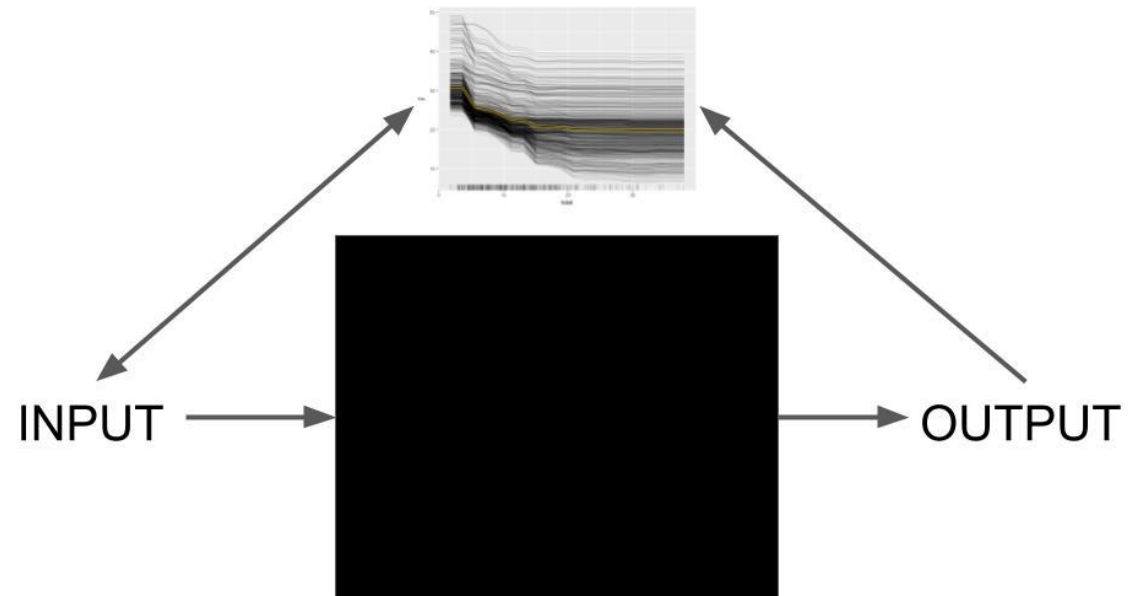
The model predictions are interpretable.

# Post-hoc interpretability

- ▶ Model-agnostic: We ignore what's inside the model and only analyze how the model output changes with respect to changes in the feature inputs. For example, permuting a feature and measuring how much the model error increases.
- ▶ Model-specific: We analyze parts of the model to better understand it. This can be analyzing which types of images a neuron in a neural network responds to the most, or the Gini importance in random forests.

# Model-agnostic post-hoc methods

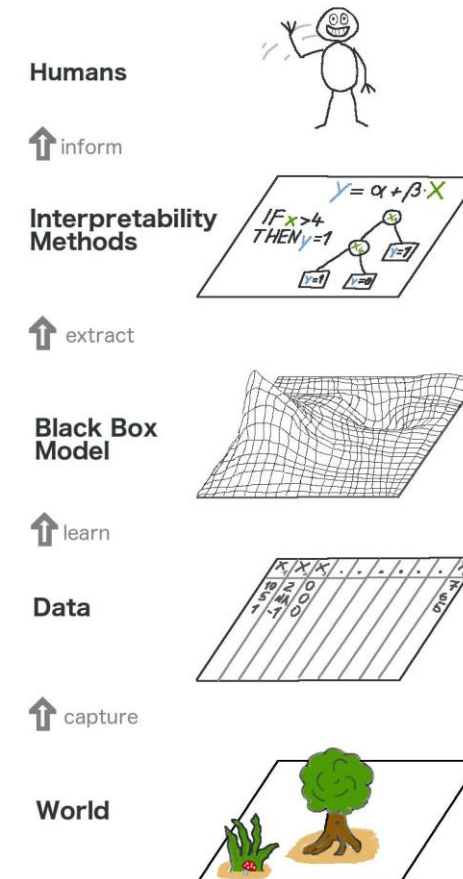
- ▶ Model-agnostic methods work by the SIPA principle: **sample** from the data, perform an **intervention** on the data, get the **predictions** for the manipulated data, and **aggregate** the results.
- ▶ Model-agnostic interpretation separates the model interpretation from the model training.





# Model-agnostic post-hoc methods

- ▶ Model-agnostic interpretation separates the model interpretation from the model training.
- ▶ Looking at this from a higher level, the modeling process gains another layer: It starts with the world, which we capture in the form of data, from which we learn a model.
- ▶ On top of that model, we have interpretability methods for humans to consume.



# Local model-agnostic post hoc methods

- ▶ Ceteris paribus plots show how changing a feature changes a prediction.
- ▶ Individual conditional expectation curves show how changing one feature changes the prediction of multiple data points.
- ▶ Local surrogate models (LIME) explain a prediction by replacing the complex model with a locally interpretable model.
- ▶ Scoped rules (anchors) are rules that describe which feature values “anchor” a prediction, meaning that no matter how many of the other features you change, the prediction remains fixed.
- ▶ Counterfactual explanations explain a prediction by examining which features would need to be changed to achieve a desired prediction.
- ▶ Shapley values fairly assign the prediction to individual features.
- ▶ SHAP is a computation method for Shapley values but also suggests global interpretation methods based on combinations of Shapley values across the data.

# Global model-agnostic methods

## Intro

Global methods describe the average behavior of a machine learning model across a dataset.

## Methods

Two broad categories within global model-agnostic methods are feature effects and feature importance.

## Feature Effects

Feature effects (PDP, ALE, H-statistic, decomposition) are about showing the relationship between inputs and outputs.

## Feature Importance

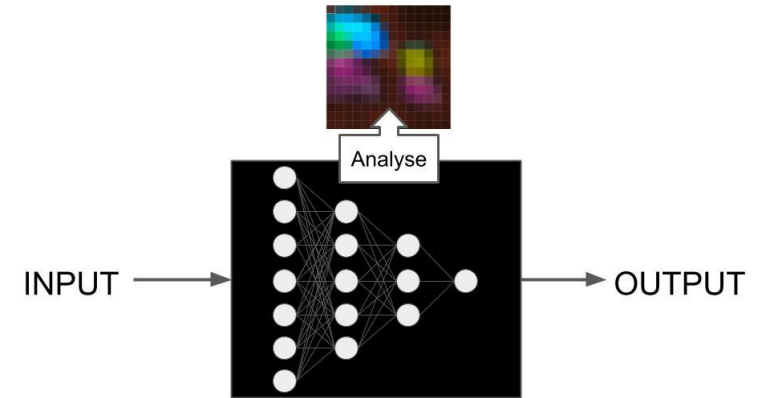
Feature importance (PFI, LOFO, SHAP importance, ...) is about ranking the features by importance, where importance is defined differently by each of the methods.

# Global model-agnostic post-hoc methods

- ▶ The partial dependence plot is a feature effect method.
- ▶ Accumulated local effect plots also visualize feature effects, designed also for correlated features.
- ▶ Feature interaction (H-statistic) quantifies the extent to which the prediction is the result of joint effects of the features.
- ▶ Functional decomposition is a central idea of interpretability and a technique for decomposing prediction functions into smaller parts.
- ▶ Permutation feature importance measures the importance of a feature as an increase in loss when the feature is permuted.
- ▶ Leave one feature out (LOFO) removes a feature and measures the increase in loss after retraining the model without that feature.
- ▶ Surrogate models replace the original model with a simpler model for interpretation.
- ▶ Prototypes and criticisms are representative data points of a distribution and can be used to improve interpretability.

# Model-specific post-hoc methods

- ▶ To make predictions with a neural network, the input data is passed through many layers of multiplication with the learned weights and through non-linear transformations.
- ▶ A single prediction can involve millions of multiplications, depending on the architecture of the neural network.
- ▶ There's no chance that we humans can follow the exact mapping from data input to prediction.
- ▶ We would have to consider millions of weights interacting in complex ways to understand a neural network's prediction.
- ▶ To interpret the behavior and predictions of neural networks, we need specific interpretation methods.
- ▶ Neural networks are an interesting target for interpretation because neural networks learn features and concepts in their hidden layers.



# Model-specific post-hoc methods

Learned Features

What features did the neural network learn?

Saliency Maps

How did each pixel contribute to a particular prediction?

Concepts

Which concepts did the neural network learn?

Adversarial Examples

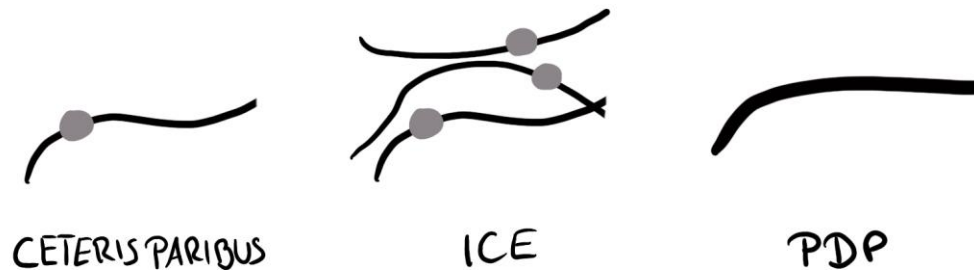
How can we fool the neural network?

Influential Instances

Extract if-then rules from data.

# Ceteris Paribus Plots

- ▶ Changing one feature but keeping the others untouched.
- ▶ It only looks at one feature at a time, systematically changes its values, and plots how the prediction changes across the range of the feature.
- ▶ By creatively combining multiple ceteris paribus curves, you can compare models, features, and study multi-class classification models.
- ▶ ICE plots are CP plots containing all CP curves for an entire dataset.
- ▶ A partial dependence plot (PDP) is the average of all CP curves of one dataset.



# Ceteris Paribus Plots

## For Numerical Features

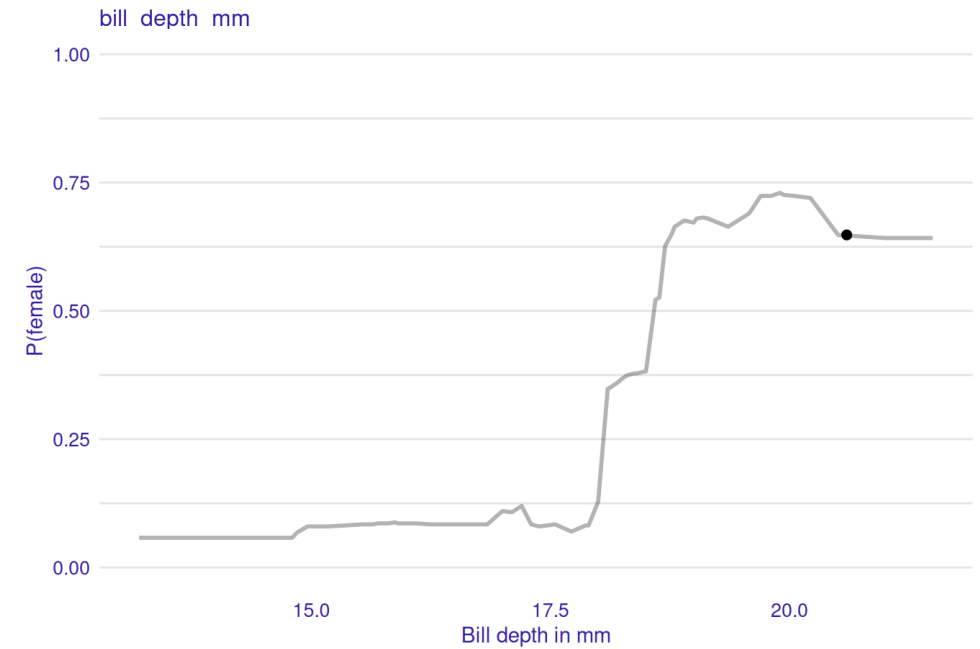


- ☐ Choose a data point and feature.
- ☐ Create a grid of feature values.
- ☐ Replace feature in the data point with grid values.
- ☐ Predict outcomes, plot value vs. prediction.

## For Categorical Features:



- ☐ List all unique categories.
- ☐ Replace original feature with each category.
- ☐ Predict and plot predictions for each.



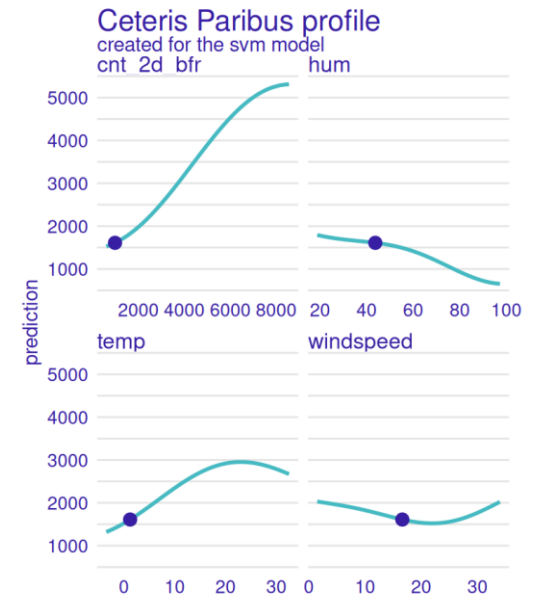
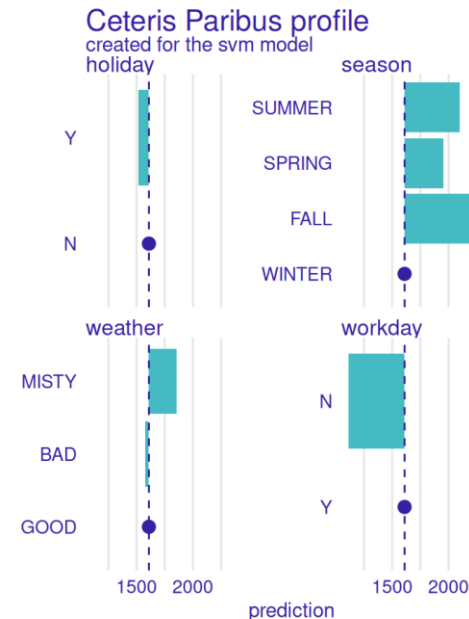


# Ceteris Paribus Plots | Example

- ▶ we look at the random forest predicting the probability of a penguin being female.
- ▶ We look at the first penguin in the test dataset, a Adelie penguin with a bill depth of 20.6 millimeters (ground truth male).
- ▶ Decreasing the bill depth of this penguin first slightly increases the predicted  $P(\text{female})$ , but then greatly decreases  $P(\text{female})$ .
- ▶ Since it's a binary classification task, but if we had more than two classes, we could plot the ceteris paribus curves for all the classes in one plot.

# Ceteris Paribus Plots | Example

- ▶ SVM predicting the number of rented bikes based on weather and seasonal information.
- ▶ We pick a winter day with good weather and see how changing the features would change the prediction.
- ▶ Changes in the number of rented bikes 2 days before would change the prediction the most.
- ▶ Also, a higher temperature would have been better for more bike rentals. Were the prediction for a non-workday, the SVM would predict fewer bike rentals.



# Ceteris Paribus Plots | Strenght & Limitations

Ceteris paribus plots are super simple to implement and understand. Attribution-based methods like SHAP or LIME don't show how sensitive the prediction function is to local changes.

They are building blocks for other interpretation methods, but you can also get creative in combining these lines across models, classes, hyperparameter settings, and features to create nuanced insights into the model predictions.

Ceteris paribus plots only show us one feature change at a time. When features are correlated, not all parts of the curve are likely or might even be completely unrealistic.

# Individual Conditional Expectation (ICE)

- ▶ Individual Conditional Expectation (ICE) plots display one line per instance that shows how the instance's prediction changes when a feature changes.
- ▶ An ICE plot (Goldstein et al. 2015) visualizes the dependence of the prediction on a feature for each instance separately, resulting in one line per instance of a dataset.
- ▶ The values for a line (and one instance) can be computed by keeping all other features the same, creating variants of this instance by replacing the feature's value with values from a grid, and making predictions with the black box model for these newly created instances.
- ▶ The result is a set of points for an instance with the feature value from the grid and the respective predictions.

# Individual Conditional Expectation (ICE)

## Centered ICE (c-ICE)

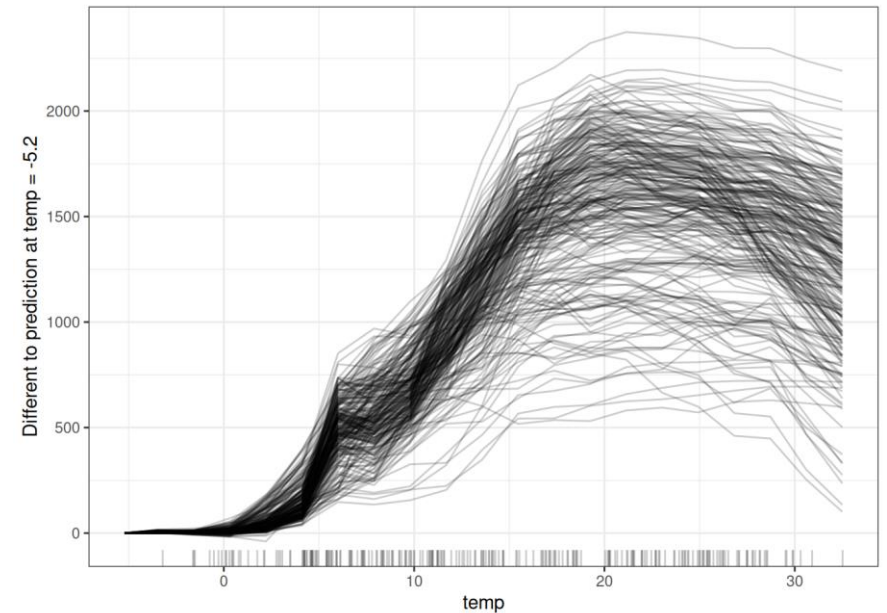


- ☐ Centers all curves at a fixed feature value.
- ☐ Shows relative prediction changes, not absolute.
- ☐ Improves comparison across instances.

## Derivative ICE (d-ICE)



- ☐ Shows rate of change (derivative) in predictions.
- ☐ Highlights interactions: if curves differ, interactions exist.
- ☐ Can include standard deviation for heterogeneity.
- ☐ Computationally expensive and less practical.



# ICE | Strength & Limitations

Replace the underlying machine learning model, you can still use the same local, interpretable model for explanation.

Using Lasso or short trees, the resulting explanations are short (= selective) & make human-friendly explanations.

LIME is one of the few methods that works for tabular data, text, and images.

The **fidelity measure** gives us a good idea of how reliable the interpretable predictions in the neighborhood of the data instance of interest.

The choice of neighborhood is an unsolved problem when using LIME with tabular data.

Data points are sampled from a Gaussian distribution, ignoring the correlation between features. This **can lead to unlikely data points**.

# Local interpretable model-agnostic explanations (LIME)

- ▶ Local surrogate models are interpretable models that are used to explain individual predictions of black box machine learning models.
- ▶ Your goal is to understand why the machine learning model made a certain prediction.
- ▶ LIME tests what happens to the predictions when you give variations of your data into the machine learning model.
- ▶ LIME generates a new dataset consisting of perturbed samples and the corresponding predictions of the black box model.
- ▶ The learned model should be a good approximation of the machine learning model predictions locally, but it does not have to be a good global approximation.

# LIME

Select Instance

Select your instance of interest for which you want to have an explanation of its black box prediction.

Perturb Dataset

Perturb your dataset and get the black box predictions for these new points.

Weight New Sample

Weight the new samples according to their proximity to the instance of interest.

Training

Train a weighted, interpretable model on the dataset with the variations.

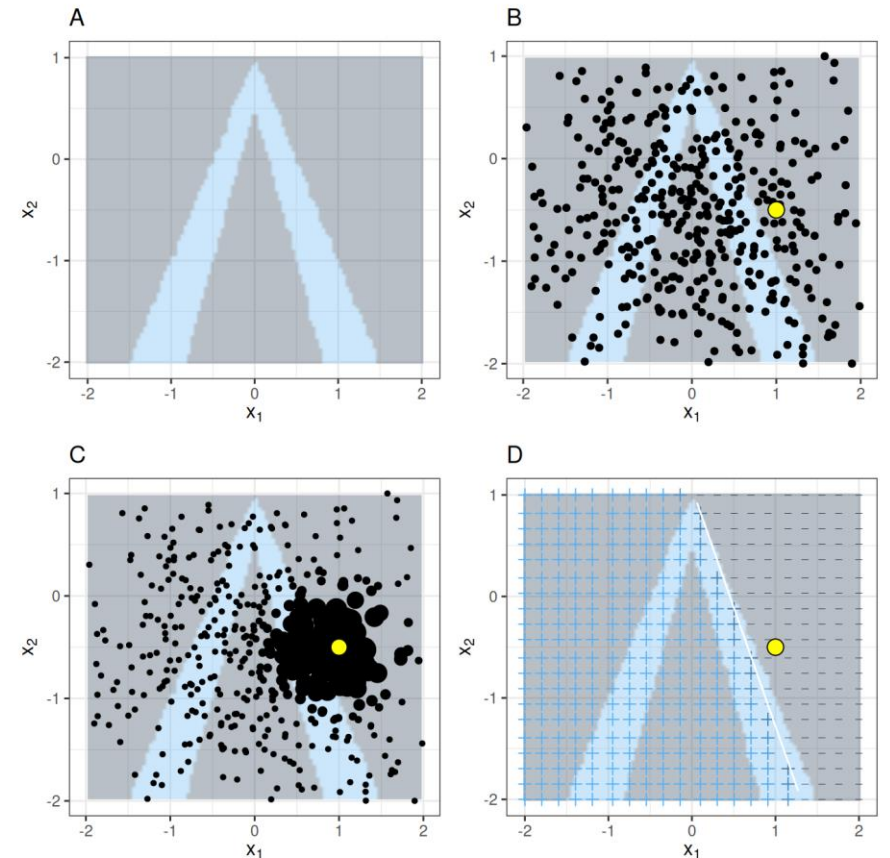
Explain

Explain the prediction by interpreting the local model.



# LIME for tabular data

- ▶ Tabular data is data that comes in tables, with each row representing an instance and each column a feature.
- ▶ LIME samples are not taken around the instance of interest, but from the training data's mass center, which is problematic.
- ▶ A smoothing kernel is a function that takes two data instances and returns a proximity measure.
- ▶ A small kernel width means that an instance must be very close to influence the local model; a larger kernel width means that instances that are farther away also influence the model.

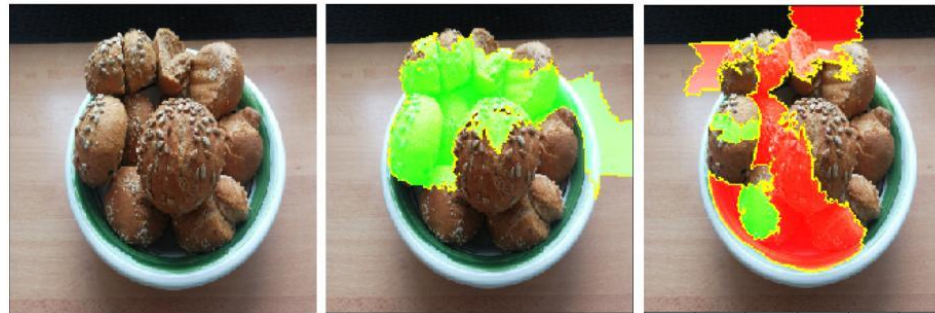


# LIME for text data

- ▶ Variations of the data are generated differently: Starting from the original text, new texts are created by randomly removing words from the original text.
- ▶ The dataset is represented with binary features for each word. A feature is 1 if the corresponding word is included and 0 if it has been removed.
- ▶ The black box model is a deep decision tree trained on the document word matrix. Each comment is one document (= one row), and each column is the number of occurrences of a given word.
- ▶ The next step is to create some variations of the datasets used in a local model.

# LIME for image data

- ▶ Intuitively, it would not make much sense to perturb individual pixels, since many more than one pixel contribute to one class.
- ▶ Randomly changing individual pixels would probably not change the predictions by much. Therefore, variations of the images are created by segmenting the image into “superpixels” and turning superpixels off or on.
- ▶ Superpixels are interconnected pixels with similar colors and can be turned off by replacing each pixel with a user-defined color, such as gray. The user can also specify a probability for turning off a superpixel in each permutation.



# LIME | Strength & Limitations

Individual conditional expectation curves are intuitive to understand. One line represents the predictions for one instance if we vary the feature of interest.



ICE curves can only display one feature meaningfully, because two features would require the drawing of several overlaying surfaces, and you would not see anything in the plot.



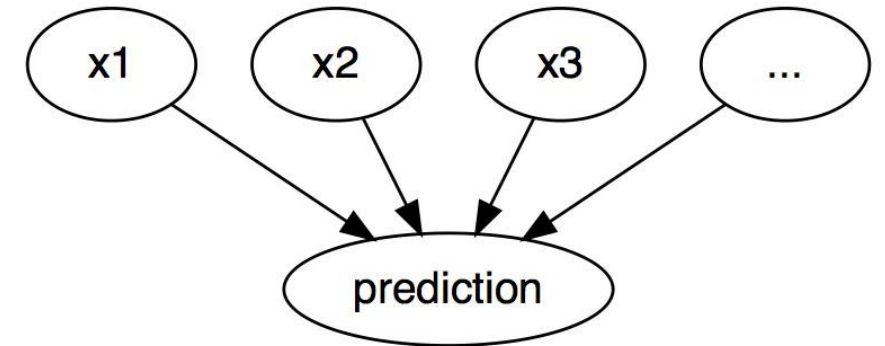
ICE curves suffer from correlation: If the feature of interest is correlated with the other features, then some points in the lines might be invalid data points according to the joint feature distribution.

# Counterfactual Explanations

- ▶ A counterfactual explanation describes a causal situation in the form: “If X had not occurred, Y would not have occurred.”
- ▶ For example: “If I hadn’t taken a sip of this hot coffee, I wouldn’t have burned my tongue.” Event Y is that I burned my tongue; cause X is that I had a hot coffee.
- ▶ Thinking in counterfactuals requires imagining a hypothetical reality that contradicts the observed facts (for example, a world in which I’ve not drunk the hot coffee), hence the name “counterfactual.”
- ▶ The ability to think in counterfactuals makes us humans so smart compared to other animals.

# Counterfactual Explanations

- ▶ We simply change the feature values of an instance before making the predictions and we analyze how the prediction changes.
- ▶ We're interested in scenarios in which the prediction changes in a relevant way, like a flip in predicted class (for example, credit application accepted or rejected), or in which the prediction reaches a certain threshold (for example, the probability for cancer reaches 10%).
- ▶ A counterfactual explanation of a prediction describes the smallest change to the feature values that changes the prediction to a predefined output.



# Counterfactual | Strength & Limitations

The counterfactual method does not require access to the data or the model. The method works also with systems that do not use machine learning. We can create counterfactuals for any system that receives inputs and returns outputs.



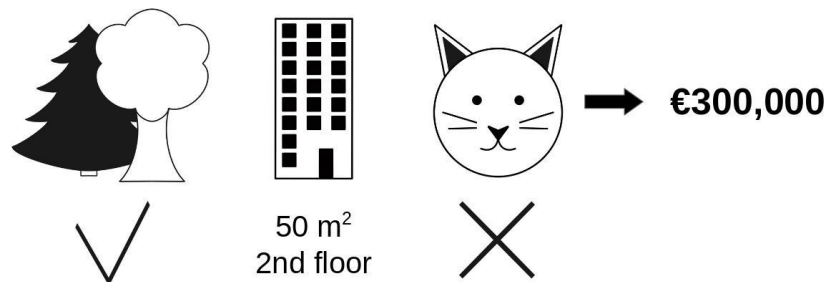
The interpretation of counterfactual explanations is very clear. If the feature values of an instance are changed according to the counterfactual, the prediction changes to the predefined prediction.



Counterfactuals are not as useful for model and data insights, since insights from counterfactual are specific to one instance and one counterfactual prediction.

# Shapley Values

- ▶ Assume the following scenario: You've trained a machine learning model to predict apartment prices. For a certain apartment, it predicts €300,000, and you need to explain this prediction. The apartment has an area of 50 m, is located on the 2nd floor, has a park nearby, and cats are banned.
- ▶ The average prediction for all apartments is €310,000. Our goal is to explain how each of these feature values contributed to the prediction. How much has each feature value contributed to the prediction compared to the average prediction?





# Shapley Values | Strength & Limitations

The difference between the prediction and the average prediction is fairly distributed among the feature values of the instance – the Efficiency property of Shapley values. LIME does not guarantee that the prediction is fairly distributed among the features. The Shapley values deliver a full explanation.



The Shapley value allows contrastive explanations. Instead of comparing a prediction to the average prediction of the entire dataset, you could compare it to a subset or even to a single data point.



In 99.9% of real-world problems, only the approximate solution is feasible. An exact computation of the Shapley value is computationally expensive because there are  $2^k$  possible coalitions of the feature values.

# SHAP

- ▶ The goal of SHAP is to explain the prediction of an instance by computing the contribution of each feature to the prediction.
- ▶ The feature values of a data instance act as players in a coalition. Shapley values tell us how to fairly distribute the “payout” (= the prediction) among the features.
- ▶ A player can be an individual feature value, e.g., for tabular data. A player can also be a group of feature values. For example, to explain an image, pixels can be grouped into superpixels, and the prediction distributed among them.
- ▶ One innovation that SHAP brings to the table is that the Shapley value explanation is represented as an additive feature attribution method, a linear model. That view connects LIME and Shapley values.